

MINI PROJECT USING JAVA

Assignment no-01

Aim -

Write a java program to showcase classes, object creation, methods, constructors using student information

Objectives -

1. To study classes, objects and creation of object of a class
2. To study methods, return a value from the method, and pass parameters to the method
3. To study constructors and use of constructors
4. To study different access specifiers in Java and their use for encapsulation.

Theory -

1. Define what is class and object with an example.

Class - is a blueprint or template that defines the properties (variables) and behaviours (method) of an object. It does not occupy memory.

Object - is an instance of a class. It represents a real-world entity and occupies memory.

Eg - class student {

int roll no;
string name;

};

2. Give the syntax/format to define methods in a class and give example for parameters passing & returning a value from methods.

Syntax -

```
accessSpecifier returnType methodName (parameter) {  
    // method body  
    return value;  
}
```

3. Write about the use of constructor with example.

A constructor is a special method use to initialize objects. It has the same name as the class & does not have a return type.

```
eg - class Student {  
    int rollno;  
    String name;  
    // constructor  
    student (int r, String n) {  
        rollno = r;  
        name = n;  
    }  
}
```

Platform -

open source Java programming tools like Eclipse Editor / Netbeans.

Conclusion -

Thus, studied what are classes, object creation, methods, constructors, access specifiers.

FAQs -

Q1. What is a class and object?

- Class - A blueprint that defines data members and methods
- Object - A real instance of a class that occupies memory

Q2 What is a constructor in Java?

A constructor is a special method that initializes an object when it is created. It has the same name as the class and no return type.

Q3 What is the default access specifier and its scope?

The default access specifier has no keyword. Its scope is within the same package only.

Q4 What are the different access specifiers?

Java has four access specifiers.

- public - accessible everywhere
- protected - accessible within package & subclasses
- default - accessible within the same package.
- private - accessible only within the same class.